PSI Center for Scientific Computing, Theory and Data

LESTO TASK 3.4 | REPORT

The LESTO case ladder, explained

How we built a simulation of where iodine sticks in a hot tube, one small, checked step at a time

Jan Kren | Paul Scherrer Institute, Villigen | 15 October 2026

Cases 001–011: Bojan Ničeno • Cases 012–024: Jan Kren

In short If a lead-cooled fast reactor ever leaks, iodine from the coolant can travel through hot gas and stick to cooler surfaces. Laboratory experiments blow such vapours through a tube whose wall runs from hot to cold and then measure *where* they land. We wrote a computer model that predicts the same thing. Instead of building it in one go, we built it as a **ladder of 24 small cases**: each one copies the previous case, changes exactly *one* thing, and comes with the numbers that prove the change works. The top rungs compare the model with real experiments: for five published PbI_2 experiments it predicts the temperature where deposits start to within 18 K on average (RMS), half the error of the two models of the original paper, with nothing tuned to the data.

1 The question

Lead–bismuth eutectic (LBE) is the liquid-metal coolant of some fast reactor designs. Fission products dissolved in it, iodine among them, can evaporate, and iodine does not leave as pure iodine: it combines with lead and bismuth into gases such as PbI_2 and BiI_3. Which compound forms matters, because each one condenses at a different temperature, and therefore ends up in a different place.

To study this, radiochemists use **thermosublimatography**: a long, narrow tube (a few millimetres wide, about a metre long) whose wall temperature falls steadily from about 1000 K at one end to near room temperature at the other. A small boat of material is heated near the hot end; a slow stream of helium carries the vapour downstream; each compound sticks to the wall where the wall is cold enough for it. Afterwards the tube is scanned, and the position of each deposit tells you its **deposition temperature**, T_{dep}.

The physics in the tube is simple to state and awkward to compute: the gas *flows* along the tube, the vapour *diffuses* sideways to the wall, and at the wall it *condenses* or *re-evaporates* depending on the local wall temperature and on what is already deposited there. The goal of the LESTO work in Task 3.4 is a computational fluid dynamics (CFD) model, built on OpenFOAM, that does all three and can be checked against the experiments.

Jargon in plain words. **Carrier**: the helium gas that flows through the tube and carries the vapour. **Frozen carrier**: we compute the helium flow once and then keep it fixed; the trace vapour is far too dilute to change it. **Mass fraction** Y: how much of the gas, by mass, is a given species. **Wall law**: the rule that says how fast vapour leaves the gas and sticks to the wall (or comes back). **HKS**: the Hertz–Knudsen–Schrage law, the textbook evaporation/condensation rate, driven by the difference between the actual vapour pressure and the equilibrium vapour pressure $p_{\mathrm{eq}}(T)$ of the solid or liquid. **Ledger / closure**: the bookkeeping of every kilogram (released, in the gas, on the wall, out of the tube); the *closure* is what is left unexplained, ideally zero. **Plug flow**: a simple hand-calculation of the same tube, used as a sanity check. **GEMS3K**: a chemical-equilibrium code that can tell the model which gases coexist.

2 The idea: a ladder, not a leap

A model like this has many ingredients: the gas flow, heat, variable gas properties, several chemical species, wall chemistry, gas-phase chemistry, and the real geometry and temperatures of each experiment. If you add them all at once and the answer is wrong, you do not know which ingredient is to blame. So we did the opposite. The folder `run/` of the LESTO repository holds 24 *cases*, numbered 001 to 024, and every case obeys the same four rules:

1. Its `readme.md` starts with *"This case is based on case N"*.
2. It changes **one thing**, and shows the changed lines.
3. It lists the **numbers to check** (e.g. "mass closure below 10^{-15}").
4. It says how to run it; cases 012–024 run with `./Allrun` and print a single word: `PASS`, `FAIL` or `NOTRUN`.

Figure 1 shows the result. The ladder climbs in four stages, which the next sections walk through: the helium flow (001–007), moving vapour through it (008–011), letting the vapour interact with the wall (012–021), and finally the comparisons with experiments and with another code (022–024). Anyone can enter at any rung, rerun it, and see the same numbers.

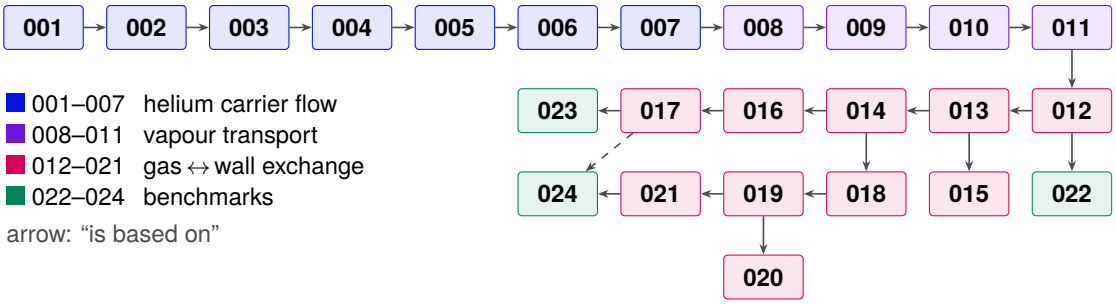


Figure 1: The 24 rungs. The trunk runs left to right along the top, then turns back along the second row. Side branches (015, 018–021, 022) add options that the main line does not need; 024 takes its chemistry from 021 and its realistic helium flow from 017 (dashed).

3 Climbing the ladder

Rungs 001–007: get the helium flowing

Bojan Ničeno

The first case is the plainest possible flow: helium through a pipe of 2.4 mm radius and 0.69 m length, at room temperature, on a mesh of 228,960 cells. Each next case adds one realistic touch: a parabolic velocity profile at the inlet (002) and inside the pipe from the start (003, which cuts the solver iterations from 388 to 222), the energy equation (004), a wall temperature that falls along the tube (005), a density that follows the ideal-gas law (006), and finally temperature-dependent viscosity and conductivity with a wall falling from 1000 K to 300 K (007, Fig. 2a). The saved result of 007 is the helium "carrier" that the next rungs read.

Rungs 008–011: move vapour through it

Bojan Ničeno

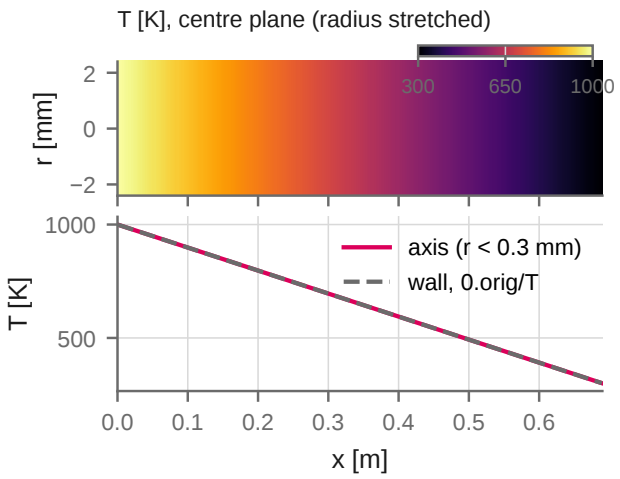
Now the flow is frozen and a new solver, `rhoFixedFlowFoam`, moves trace species through it. Case 008 sends PbI_2 vapour and an inert tracer with identical settings and checks that they stay identical (they do, to the last digit). 009 gives PbI_2 its real, temperature-dependent diffusivity, up to 25 times the tracer's on the hot side. 010 introduces a *solid* species, the deposit, and the interface through which chemistry will later talk to the flow. 011 couples them: what leaves the gas appears as solid on the wall, and the two sources add up to exactly zero (Fig. 2b).

Rungs 012–021: teach the vapour how to stick

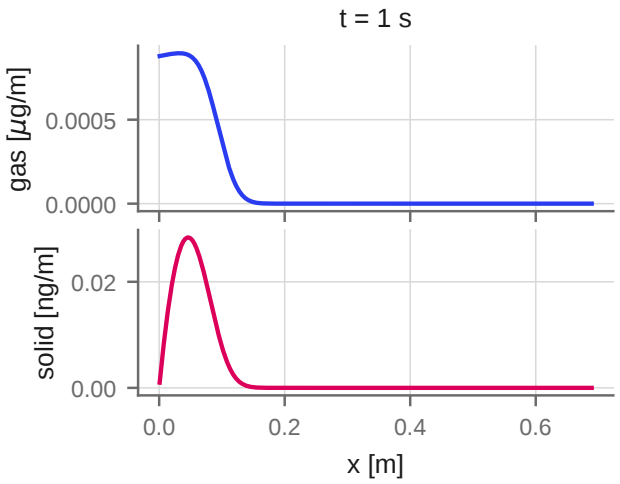
Jan Kren

Up to 011 the deposition rule was a placeholder. From 012 on, the cases add real wall physics, again one piece at a time:

- **012** The simple *temperature model* of Lobresco et al. [1]: vapour sticks once the wall is colder than a set T_{dep}. It also introduces a vapour source (a "boat") and the **mass ledger**: every step, every kilogram is accounted for. The unexplained remainder stays at about 10^{-16} of the released amount, i.e. at the limit of computer arithmetic (Fig. 3a).
- **013** The physical wall law (HKS): condensation and re-evaporation driven by the equilibrium vapour pressure of solid and liquid PbI_2, and a boat that evaporates its real inventory.
- **014** A cheap synthetic test channel (1000 K to 350 K) that runs in 10 s. Here the model's T_{dep} can be checked against the plug-flow hand calculation: 561.2 K against 567.1 K, within the 10 K gate (Fig. 3b).



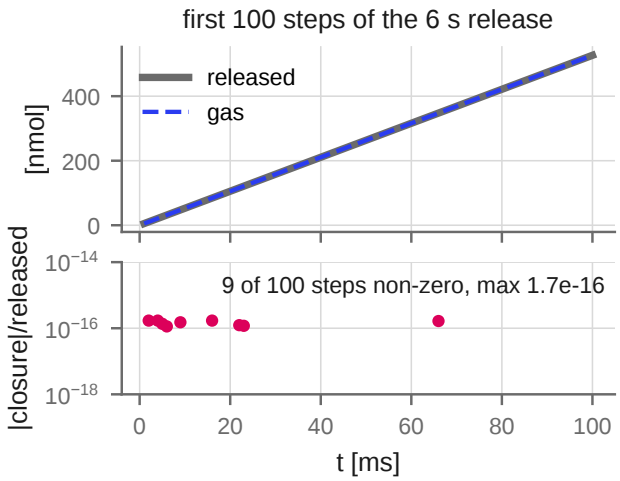
(a) 007: the helium carrier. Top: temperature in the tube (radius stretched); bottom: the gas on the axis follows the wall from 1000 to 300 K.



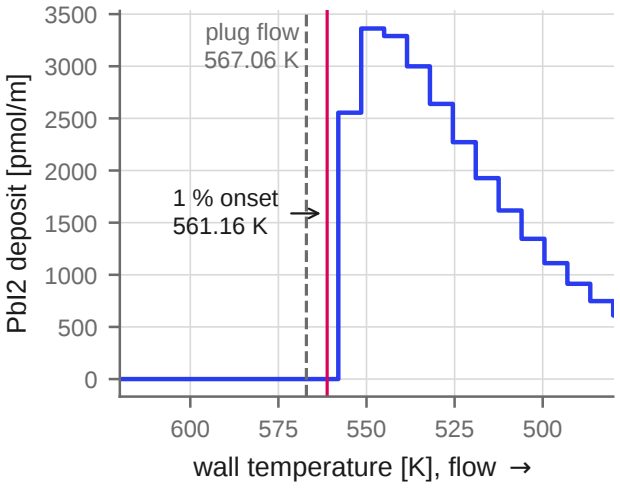
(b) 011: after 1 s, PbI_2 gas enters from the left (top) and turns into deposit on the wall (bottom), mass for mass.

Figure 2: The first two stages: a correct carrier, then vapour that moves and deposits. (a) is drawn from the saved carrier of 007; (b) is a rerun of 011 to 1 s, made on 15 October 2026.

- **016** The vapour has to *diffuse* to the wall before it can condense; this extra resistance turns out to control the uptake. It is used in every realistic run after this one.
- **017** The real helium flow and measured wall temperature of one Lobresco experiment, instead of the synthetic channel.
- **018–021** More chemistry: a second iodide (BiI_3), nine Pb–Bi–I gases that re-balance with each other everywhere in the tube (019; 020 does the same with GEMS3K), and metal vapour from LBE with shared deposits and wall reactions (021). In 021, all five deposit onsets land within 3.9 K of the hand calculation.



(a) 012: released and gas inventory (top) match; the unexplained remainder (bottom) is $\sim 10^{-16}$, machine round-off.



(b) 014: PbI_2 deposit along the synthetic channel. The model's onset (1 % rule, 561.2 K) sits 5.9 K from plug flow.

Figure 3: Two kinds of check used throughout the ladder: nothing is lost (a), and the answer agrees with a hand calculation where one exists (b).

4 The top rungs: does it match reality?

The last three cases do not add physics; they *test* it.

022 — against another code. Lobresco et al. computed their experiments with the code T-Flows using the simple temperature model. In "paper mode" our model repeats their calculation: the gas level agrees with their Fig. 9 to 0.5 %, and all ten lines of their Table 2 agree within the resolution of the wall temperature table. The

one visible difference, a deposit starting 2.6 cm earlier, disappears (to 0.9 mm) with the T-Flows parameter set; which set made the published figures is a question for the authors.

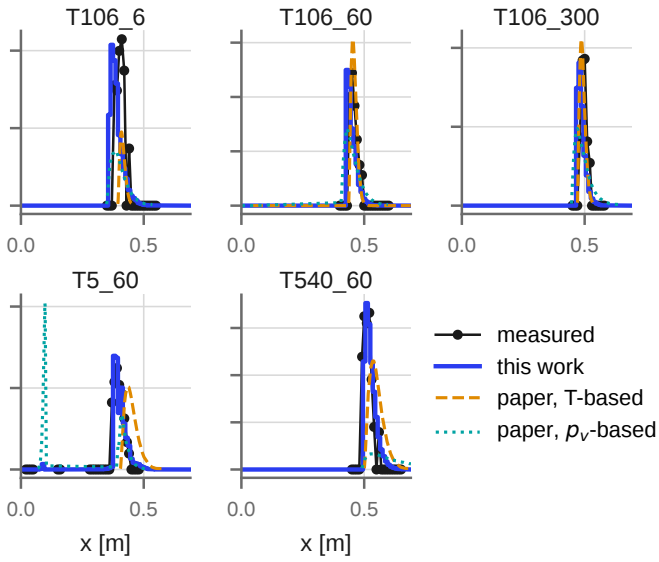
023 — against five experiments. The five PbI_2 experiments of Lobresco et al. [1] differ in helium flow (5, 106 and 540 mL/min) and duration (6 to 300 s). We ran all five with the HKS wall law, the measured wall temperatures, and *no parameter adjusted to the data*, then applied the paper's own rule for T_{dep} (the point where the deposit reaches 1 % of its peak). Table 1 and Fig. 4a show the outcome: an RMS error of 18 K, against 35 K for both models of the original paper. A coarser mesh changes the answers by at most 0.3 K, and the mass ledger closes to 3.3×10^{-16}.

Table 1: Case 023: predicted minus measured deposition temperature, production mesh. Experiment names: helium flow (mL/min) _ duration (s).

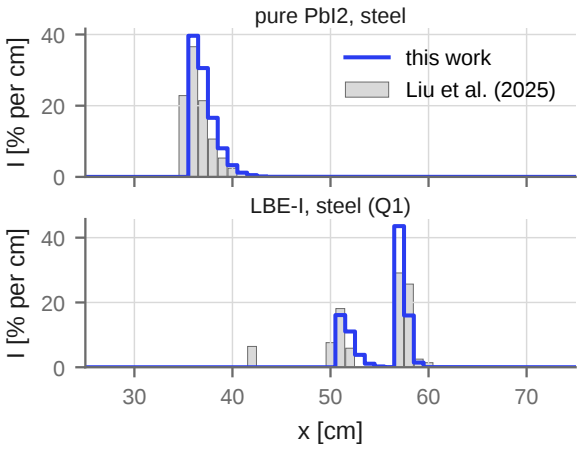
	T106_6	T106_60	T106_300	T5_60[a]	T540_60	**RMS**
This work (023)	+26 K	+19 K	+16 K	−11 K	−14 K	**18 K**
Lobresco et al., both models						35 K

[a] At 5 mL/min some vapour diffuses upstream and leaves a small deposit near the cold inlet; the value is for the main deposit downstream of the boat.

024 — the full Pb–Bi–I chemistry. Liu et al. [2] evaporated pure iodides and iodine-containing LBE into helium in steel and silica columns. Case 024 runs their configurations with the full chemistry of 021 on the realistic flows of 017, 60 s each on a mesh of 41,472 cells. For pure PbI_2 on steel the computed peak sits at 36 cm, 366.5 °C, against the measured 368 ± 18 °C, and for iodine from LBE the model reproduces the two separate deposit bands (Fig. 4b). Refining the mesh to 93,312 cells moves the profiles by 1.98 %.



(a) 023: deposit profiles of the five Lobresco experiments (arbitrary units) against tube position.



(b) 024: iodine deposit per cm, pure PbI_2 (top) and iodine from LBE (bottom), against Liu et al. (2025).

Figure 4: The benchmarks. Black dots and grey bars: measurements. Blue: this work. Dashed and dotted: the two models of the original paper.

5 What this gives us, and what it does not

What we have

- **A model you can trust step by step.** Every capability of the solver has a rung that isolates it and a number that checks it. When something breaks later, `./Allrun-ladder 012-024` reruns all checks and points at the first rung that fails.
- **Nothing goes missing.** In every exchange case the bookkeeping of lead, bismuth and iodine closes at 10^{-16}, the limit of double-precision arithmetic.

- **A useful prediction.** Against five experiments the untuned model halves the error in T_{dep} of the published models (18 K vs 35 K RMS), and it reproduces the multi-band deposits of the LBE experiments.
- **A paper trail.** Cases 023 and 024 feed directly into the methods manuscript (Sec. 6.3, Table V, Fig. 5) and the ladder is the availability section of LESTO deliverable D38.

What we do not have yet

- Some inputs of the experiments are not published and are *assumed*: how fast the boat released its material, the reference conditions of the flow, and several positions. The 024 results are therefore labelled provisional.
- The model's deposit peaks sit 1 to 4 cm upstream of the measured ones in 023; the boat's obstruction of the flow and the dense vapour right next to it are not represented.
- The BiI_3 cases (018, 019, 021) need restricted thermodynamic tables that cannot be redistributed, and 015 and 020 need an optional GEMS3K build; without them those cases print `NOTRUN`.
- The mesh study of 024 shows small changes (1.98 %) but no formal grid-convergence index: the apparent orders of convergence are not consistent.

How to try it Every case lives in `run/<NNN-name>/`. Read its `readme.md`, then run `./Allrun` (cases 012–024; outputs go into a git-ignored `work/` folder) and look for `PASS`. The cheap cases take seconds to minutes; 017 about 8 minutes; the full 023 and 024 studies take hours per experiment. The slide-by-slide version of this report is `presentations/case-ladder/lesto-case-ladder.pdf`.

References

[1] F. Lobresco, G. Tiebel, G. C. Conigliaro, P. Steinegger, O. Marinich, N. I. Prasianakis, J. Neuhausen, B. Ničeno, *Assessment of temperature-based and vapor-pressure-driven models for CFD simulation of volatile species deposition*, Nucl. Eng. Des. 459 (2026) 115201. doi:10.1016/j.nucengdes.2026.115201

[2] L. Liu, V. Zobnin, I. Zivadinovic, R. Eichler, A. Vögele, D. Herrmann, A. Aerts, P. Steinegger, J. Neuhausen, *Chemical speciation of iodine evaporated from liquid lead–bismuth eutectic investigated by thermosublimatography*, J. Radioanal. Nucl. Chem. 334 (2025) 5201–5216. doi:10.1007/s10967-025-10246-4